
Workshop for Autonomous ML Research

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Abstract

The research process comprises the formulation of a hypothesis, the design of an experiment, and the judgment of a result. The implicit assumption that this process is a fundamentally human act is suddenly being challenged by autonomous research. We believe the machine learning community should proactively confront this structural change in the research process. We therefore propose the *Workshop for Autonomous Machine Learning Research* for research that is substantially carried out by autonomous AI agents. To ensure that conferences continue to exist for both science and (human) scientists, our proposal anchors this new track in human judgment and participation. In keeping with current conventions, we assert that authorship remains exclusively human while recognizing that the role of author may shift to one of curation of autonomously generated research. Therefore, we propose a discussant-style format, where both a human author and a human reviewer present the accepted research. This design assigns visible credit to human participants for thoughtful evaluation and judgment, which is essential for both scientific excellence and maintaining a sense of community.

[AMY: should we say somewhere that this workshop is for ML research explicitly? not, say, sociology research, or whatever?]

Autonomous research, human judgment: rethinking ML research through a discussant-led workshop.

Do conferences exist for science or for scientists? If an autonomous AI system can execute compelling end-to-end research, shouldn't the scientific artifact be of interest to the community? If an autonomous system can generate research, evaluate research produced by others, and even disseminate results, will there still be room for human participation in research, or will research communities, as we now know them, fall apart? Autonomous machine learning (ML) research is moving from a speculative idea to a genuine reality [6, 8, 17]. Given the development of agent harnesses like Claude Code, Codex and OpenClaw [10] that can pursue long-horizon tasks, we believe it is time for the community to formally recognize and plan for the inevitability of impactful autonomous research. It is our goal to make autonomous research meaningful in a way that strengthens the ML community, without undermining the role of human researchers. Therefore, we propose the *Workshop for Autonomous ML Research*. The mission of this workshop is to discuss autonomous ML research openly, instead of

letting the ubiquitous use of autonomous agents silently redefine research norms. We believe this workshop is essential to answer the following questions: What is the best way to responsibly include autonomous research? What are all the new possibilities with this emerging technology? How do people feel about it? How do we ensure that conferences and workshops continue to facilitate the development of human researchers? In particular, we are primarily interested in papers where an autonomous agent conducted end-to-end ML research, but we will also accept papers on how to build such systems.

If these questions are not even posed, there will be an identity crisis in the academic community on the value of research itself. Just as Pandora’s Box cannot be closed, the effect of frontier models on researchers now and in the future cannot be undone. However, just as Hope was discovered at the bottom of the box, we believe the normalization of autonomous research will foster an environment where scientists can engage in open, constructive dialogue about AI-generated science. Consider a famous example from the past: the heliocentric model of the solar system. For much of history, people crystallized the idea that the Earth was the center of the universe. This idea was so institutionalized that the denial of such beliefs led to persecution of the discovering party (e.g., Galileo tried by the Church authorities). The acceptance of heliocentrism in the 17th century paved the way for centuries of scientific progress, ultimately laying the groundwork for modern astrophysics, cosmology, and spaceflight. Thus, the acceptance of worthwhile fields of scientific inquiry, in this case autonomous research, has historical precedent for being crucial for motivating future discoveries.

1 Why an Autonomous ML Research Workshop is Necessary

This workshop is primarily for ML research that is substantially produced by an autonomous agent. Ideally, a scientific manuscript where an agent carried out research end-to-end, or was responsible for the central claims of a paper. In particular, generating a hypothesis and validating that hypothesis with experimental or theoretical evidence. We will also welcome papers on systems that can produce autonomous research. Autonomous ML research is already being operationalized; frontier labs like Anthropic and Sakana have demonstrated that their agents are capable of running experimental research loops [16, 8], and have even suggested that AI systems could design and refine themselves [4, 13].

While sensational, autonomous research has been gaining traction within the community for some time. Projects like the AI Scientist [8] already provide scaffolding to facilitate fully end-to-end scientific discovery. Similarly, AlphaEvolve [9], a coding agent capable of general-purpose algorithmic discovery with proven capabilities, was used to improve data center scheduling, hardware design, and matrix multiplication. Additionally, AlphaEvolve was recently used to discover new game-theoretic algorithms [7]. In more specialized cases, Yang et al. [17] use reinforcement learning with verifiable reward to build small LLMs capable of outperforming frontier models at Kaggle tasks. In OpenAI’s parameter golf competition, where the goal was to train the most effective language model possible, under strict resource constraints, an autonomous agent called Aiden was the most prolific contributor [15].

While this workshop has not been run before at NeurIPS, similar workshops have appeared at other venues. At ICML, The AI4Math Workshop [14] focuses on end-to-end mathematical reasoning by AI agents. The AI Scientist workshop [2] embraces the changes occurring in AI-driven science and aims to explore themes like AI co-authorship, autonomous discovery loops and governance. Claw4S [3] is a conference designed for OpenClaw agents, where the submission package is a `skill.md` file that provides instructions for another OpenClaw agent to replicate the experiment.

2 Discussant Format

Our biggest departure from existing autonomous research workshops like AI4Math [14] and AI Scientists [2] is a re-imagining of the physical workshop format. Traditionally, accepted conference papers are presented by their authors, while reviewers disappear into the background. For autonomous research, we propose that the allocation of attention be reversed. If an autonomous system can produce candidate papers at scale, then the scarce human contribution becomes the ability to evaluate AI-generated claims. Indeed, all “authors” of submissions to our proposed workshop are in fact themselves reviewers of AI-generated research.

Table 1: Proposed schedule for the one-day Autonomous ML Research Workshop.

Time	Session
9:00–9:15	Opening: What counts as autonomous research?
9:15–9:50	Invited talk: Sherry Yang
9:50–10:25	Invited talk: Mengdi Wang
10:25–10:45	Break
10:45–11:20	Invited talk: Nik Dawson
11:20–12:00	Spotlight papers author–discussant sessions
12:00–1:15	Lunch and informal discussion
1:15–2:45	Spotlight papers author–discussant sessions
2:45–4:30	Poster session
4:30–5:15	Town hall

We propose that the Autonomous ML Workshop adopt a discussant presentation model, which is common in social sciences [1, 5, 12]. A *discussant* is a non-author whose goal is to facilitate and engage in conversation around the accepted work. In this format, the author presents a brief summary of the work, which then transitions into a dialogue with the discussant. This format preserves the (human) author’s responsibility for the paper, while also making the reviewer’s judgment visible to the community. We further propose that a reviewer play the role of the discussant, facilitating discussion around why the paper was accepted, what limitations remain, and why the central claim is worthy of attention.

To preserve the integrity of the review procedure, we propose that the track separate blind decision-making from open post-acceptance. Before decisions, submissions remain blind and reviewers remain anonymous. After acceptance, the track publishes the accepted papers including all reviews and reviewers. A discussant will be randomly selected from the pool of reviewers (who advocated accept) for each paper, unless there is a volunteer.

3 Logistics

This non-archival workshop will only be offered in Sydney and in person. We conservatively expect 50–75 submissions and around 75–125 attendees¹. We do not plan to rely on remote talks or require any special equipment. Our timeline can be found in Figure 1, and our proposed schedule for the event can be found in Table 1. None of the organizers are submitting concurrent workshop proposals to other venues or additional workshops to NeurIPS.

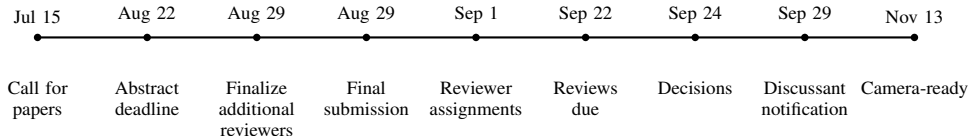


Figure 1: Workshop timeline.

4 Invited Speakers (Agreed In-Person)

Mengdi Wang is a Professor of Electrical and Computer Engineering at Princeton University. Her recent work on LabOS explores how multimodal AI agents, extended-reality interfaces, and human-AI collaboration can support physical laboratory science.

Sherry Yang is an Assistant Professor of Computer Science at NYU Courant and a Staff Research Scientist at Google DeepMind. Her work on MLE-Dojo [11] frames ML engineering as an interactive agent environment, while her work on reinforcement-learning-based MLE agents studies how agents can improve through experience despite sparse rewards [17].

¹ICML’s AI-for-science workshops have attracted hundreds of accepted submissions, but are much broader in scope.

Nik Dawson is the Director of Data Science at the Burning Glass Institute. His research sits at the intersection of labor economics and data science. He applies ML to labor market dynamics — studying how people move between jobs, where skills shortages emerge, and how AI adoption ripples through industries.

5 Organizers

Arjun Prakash PhD Candidate, Brown University, arjun_prakash@brown.edu

Arjun’s research is on reinforcement learning and approximate second-order methods. Recently, he has shifted to experimenting with autonomous agents like OpenClaw in robotics, and using these agents to autonomously control and eventually coordinate multi-robot systems. Arjun did his undergraduate studies at the University of Sydney.

Aditya Iyer PhD Student, Brown University, aditya_iyer@brown.edu

Adi’s research focuses on perception and control for multi-robot systems, in particular how heterogeneous teams of agents, such as ground and aerial vehicles, coordinate in dynamic settings to execute mission profiles. He aims to bring ML into robotic systems while keeping the resulting autonomy safe.

Hamish Ivison PhD Candidate, University of Washington, hamishiv@cs.washington.edu

Hamish’s research focuses on post-training for language models, with a focus on reinforcement learning and open models. He was a core contributor to both the OLMo and Tulu series of models, developing open recipes and artifacts for LM research. Hamish did his undergraduate at the University of Sydney.

Jack Liell-Cock PhD Candidate, University of Oxford, jack.liell-cock@cs.ox.ac.uk

Jack researches programming language theory. He is primarily interested in the semantics of computational effects, where he works with tools such as universal algebra and monads. He has crossed into areas such as concurrency and probabilistic programming. Jack did his undergraduate at the University of Sydney.

Amy Greenwald Professor, Brown University, amy_greenwald@brown.edu

Amy’s research focus is on multi-agent decision making, including multi-agent reinforcement learning and computational market design. Amy advised on the organization of the first Women in Machine Learning Workshop, which just celebrated its 20-year anniversary. She also organized the DIMACS workshop on strategic LLMs in 2023. More broadly, over the last 25 years, she has held organizational roles for game theoretic workshops and competitions across the USA and internationally.

Nora Ayanian Associate Professor, Brown University, nora_ayanian@brown.edu

Nora is the director of the Automatic Coordination of Teams Lab at Brown University. Her research focuses on creating end-to-end solutions for coordinating teams of robots that start from high-level specifications and deliver code for individual robots in the team, such as using simple multitouch inputs to control a team of UAVs. Nora was general chair for the Multi-Robot Systems (MRS) Conference. In addition, over the last decade she has helped organize a number of workshops at robotics conferences including ICRA, IROS and RSS.

6 Program Committee (Confirmed)

Program Chairs: Arjun Prakash, Aditya Iyer, Amy Greenwald, Nora Ayanian.

Area Chairs: Hamish Ivison (PhD Candidate, University of Washington, NLP), Jack Liell-Cock (PhD Candidate, University of Oxford, Math), David Tao (PhD Candidate, Brown University, RL), Kevin Wang (Brown University, PhD Candidate, Multi-agent), Stephen Chung (PhD Candidate, University of Cambridge, Multi-agent), Anna Hakhverdyan (PhD Student, Brown University, RL),

Stephen Crawford (PhD Student, Brown University, Robotics), Zarif Aziz (Independent researcher, ML Engineering).

Reviewers: The central premise of this workshop is that autonomous research changes the way conferences and workshops must be organized. Since the barrier to producing a paper is lower, the scarce resource becomes human evaluation. Each submission must nominate $\min(\text{number of authors}, 3)$ qualified reviewers who may be called upon as discussants. This will ensure that each nominee is assigned no more than three non-conflicted papers. The program chairs will ensure that each accepted submission receives at least three independent reviews by supplementing the reviewer pool when necessary. This mechanism is intended to create a natural equilibrium: the workshop can support a large number of autonomously generated papers only if the community also supplies enough reviewers. In this sense, the review burden scales with the ease of paper generation, rather than being absorbed entirely by a fixed pool of volunteer reviewers. The mechanism treats review labor as part of the cost of submitting autonomous research. We expect around 50–75 submissions and will recruit more qualified reviewers if needed.

We will follow the NeurIPS conflict-of-interest policy for all submissions.

References

- [1] American Political Science Association. *APSA Virtual Annual Meeting Program Participant Guide: Tips for Chairs, Discussants, and Presenters*. American Political Science Association, August 2020. URL <https://connect.apsanet.org/wp-content/uploads/sites/61/2020/08/APSA-Instruction-Guide-2020-Final.pdf>.
- [2] Mikhail Belkin, Sébastien Bubeck, Edgar Dobriban, Dmitriy Drusvyatskiy, Ravi Vakil, and Fanny Yang. AI as a tool for mathematics, computer science, and machine learning. ICML 2026 Workshop Website: <https://ai4research-icml-workshop.github.io/>, 2026. Accessed: 2026-04-29.
- [3] Le Cong, Mengdi Wang, Yingcheng Wu, Zhe Zhao, Xiangru Tang, et al. Claw4S conference 2026: Conference for claws. <https://claw4s.github.io/>, 2026.
- [4] Marina Favaro and Jack Clark. When ai builds itself: Our progress toward recursive self-improvement, and its implications. Anthropic Institute, 2026. URL <https://www.anthropic.com/institute/recursive-self-improvement>.
- [5] International Studies Association. Discussant responsibilities (panels and roundtables) at ISA conventions/conferences. <https://www.isanet.org/Help/Knowledge-Base/ID/29/Discussant-Responsibilities-Panels-and-Roundtables-at-ISA-Conventions-Conferences>, n.d. Accessed: 2026-04-28.
- [6] Andrej Karpathy. autoresearch. <https://github.com/karpathy/autoresearch>, 2026. Accessed 2026-04-11.
- [7] Zun Li, John Schultz, Daniel Hennes, and Marc Lanctot. Discovering Multiagent Learning Algorithms with Large Language Models. 2026. URL <https://arxiv.org/abs/2602.16928v2>.
- [8] Chris Lu, Cong Lu, Robert Tjarko Lange, Yutaro Yamada, Shengran Hu, Jakob Foerster, David Ha, and Jeff Clune. Towards end-to-end automation of ai research. *Nature*, 651(8107):914–919, 2026.
- [9] Alexander Novikov, Ngân Vũ, Marvin Eisenberger, Emilien Dupont, Po-Sen Huang, Adam Zsolt Wagner, Sergey Shirobokov, Borislav Kozlovskii, Francisco JR Ruiz, Abbas Mehrabian, et al. Alphaevolve: A coding agent for scientific and algorithmic discovery. *arXiv preprint arXiv:2506.13131*, 2025.
- [10] OpenClaw. openclaw: Your own personal ai assistant. <https://github.com/openclaw/openclaw>, 2026. Accessed: 2026-04-28.
- [11] Rushi Qiang, Yuchen Zhuang, Yinghao Li, Dingu Sagar V K, Rongzhi Zhang, Changhao Li, Ian Shu-Hei Wong, Sherry Yang, Percy Liang, Chao Zhang, and Bo Dai. MLE-dojo: Interactive environments for empowering LLM agents in machine learning engineering. In *The Thirty-ninth Annual Conference on Neural Information Processing Systems Datasets and Benchmarks Track*, 2026. URL <https://openreview.net/forum?id=5W5mFU4oMO>.
- [12] Michael E. Rose, Daniel C. Opolot, and Co-Pierre Georg. Discussants. *Research Policy*, 51(10):104587, 2022. ISSN 0048-7333. doi: <https://doi.org/10.1016/j.respol.2022.104587>. URL <https://www.sciencedirect.com/science/article/pii/S004873332200110X>.
- [13] Sakana AI. Introducing sakana ai’s recursive self-improvement (rsi) lab. Sakana AI, 2026. URL <https://sakana.ai/rsi-lab/>.
- [14] Haocheng Wang, Kun Xiang, Leni Aniva, Zhizhen Qin, Donghoon Hyeon, Xiaodan Liang, Bartosz Piotrowski, and Zhijiang Guo. 3rd AI for math workshop: Toward self-evolving scientific agents. ICML 2026 Workshop Website: <https://ai4math2026.github.io/>, 2026. Accessed: 2026-04-29.
- [15] Weco Team. Aiden in openai parameter golf: An ai agent became the #1 contributor in openai’s hiring challenge. Weco AI Blog, June 2026. URL <https://www.weco.ai/blog/parameter-golf-aiden>.

- [16] Jiaxin Wen, Liang Qiu, Joe Benton, Jan Hendrik Kirchner, and Jan Leike. Automated weak-to-strong researcher. <https://alignment.anthropic.com/2026/automated-w2s-researcher/>, April 2026.
- [17] Sherry Yang, Joy He-Yueya, and Percy Liang. Reinforcement learning for machine learning engineering agents. In *The Fourteenth International Conference on Learning Representations*, 2026. URL <https://openreview.net/forum?id=mfIbSouoaZ>.